# The Role of Water in the Superconductivity of Sodium Cobaltate

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Sodium cobaltate superconducts only when it is wet, and after twenty years there is no agreed reason why. We take the water seriously as a mechanical object and calculate, by ordinary density-functional methods, what happens between the $CoO_2$ sheets when the gallery that holds the sodium is pulled open. Past a critical spacing two things happen at once: the charge on the oxygen layers bifurcates, so an electron gas condenses in the gallery, and the stiff sodium well splits into a shallow double well in which the ion begins to rattle. The gas supplies carriers and the rattle supplies glue, and a phonon-mediated superconducting dome sits in the narrow window between the layers too close, where there is no gas, and the sodium frozen off-centre, where the coupling runs away into a polaron. The dry parent and the monolayer hydrate fall correctly outside this window. One number does not fit: the superconducting bilayer hydrate holds its layers wider than the static window, and resolving that mismatch forces the paper's one real claim about the water. The water is not a spacer. It is a lubricant, and that statement can be tested.

## I. INTRODUCTION

It is a curious fact that sodium cobaltate becomes a superconductor only when it is wet. Dry $Na_xCoO_2$ is an ordinary metal down to the lowest temperatures measured; slip two layers of water between its $CoO_2$ sheets and it superconducts near 4.5 K [1]; and twenty years of looking for the reason inside the $CoO_2$ sheet itself have not produced an accepted mechanism [2]. This paper takes the water seriously as a mechanical object and asks what it does. We shall show, by ordinary density-functional calculations, that pulling the layers apart past a critical separation does two things at once: the charge on the oxygen layers bifurcates, so that an electron gas condenses in the gallery between the sheets, and the sodium ion, which had been sitting at the bottom of one stiff well, finds itself with two shallow wells and begins to rattle between them. The gas supplies the carriers and the rattle supplies the glue, and superconductivity can live only in the window between the layers too close, where there is no gas, and the sodium frozen, where there is no glue (Fig. 1).

There is nothing exotic in the calculations, and the ingredients—a two-dimensional electron gas, an anharmonic mode—are old ones; what is new is only where we look for them, which is between the layers rather than in them. The electronic background is the one worked out by Radha and Lambrecht for the surfaces of the layered cobaltates [3], whose surface bands and their spin polarization are the seed of everything below; the present paper is its sequel, moved from the surface into the gallery.

One measured number does not fit this static picture. The superconducting bilayer hydrate holds its layers 9.9 Å apart, wider than the window we calculate, and there the static calculation says the sodium is not rattling at all but bound hard against one $CoO_2$ layer. We shall meet this difficulty early, in Sec. IV, because resolving it is what teaches us the actual role of the water. The water is not a spacer. It is a lubricant.

We first show what the calculations find when the gallery opens, told as a single event seen three ways. We then build the electron–phonon model that turns that event into a superconducting dome and locate its two walls. We take up the 9.9 Å difficulty in its own section. Finally we ask what the picture must then explain—why lithium never superconducts, where the magnetic phase sits, what a valence probe should read—and where the calculation is weakest.

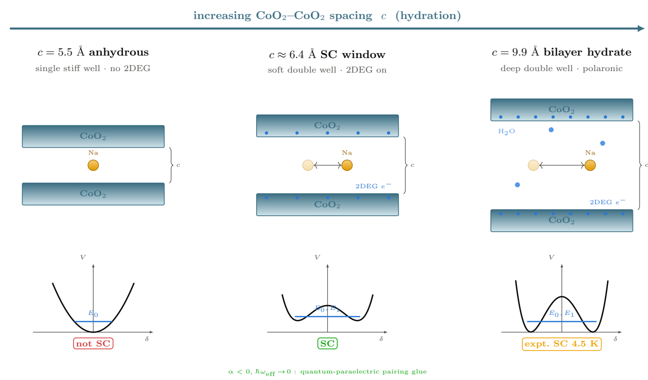


FIG. 1. The mechanism in one picture. When the gallery between the $CoO_2$ sheets is narrow (left), the sodium sits at the centre of a single stiff well and the electrons stay bound to the oxygen: no gas, no soft mode, no pairing. Open the gallery past a critical spacing (right) and both change at once—charge spills back into the gallery as a two-dimensional electron gas, and the sodium well splits into a shallow double well in which the ion rattles. The gas carries and the rattle glues. Superconductivity lives in the narrow window between these two states and the polaronic state past it.

## II. WHAT HAPPENS WHEN THE GALLERY OPENS

Picture the sodium ion sitting in the gallery, midway between two $CoO_2$ sheets, and slowly pull the sheets apart. While the gallery is narrow the ion sits at the centre in a single stiff well; there is one lowest-energy place for it and it stays there. Open the gallery far enough and the centre stops being the bottom of the well and becomes the top of a small barrier: the ion now has two equivalent places to sit, one toward each sheet, and between them it rattles. At the same spacing, and this is the point, the electrons do something of their own. The charge that the oxygen layers had been holding splits, and part of it spills back into the gallery as a two-dimensional electron gas. One event, two faces.

We see this in the total energy directly. For each spacing $c$ we move the sodium off centre by a displacement $\delta$ and record the energy $E(\delta)$; the calculations are spin-polarized PBE with projector-augmented waves, at fixed in-plane geometry, and are described in Sec. VI. Near the bottom the curve is even in $\delta$, so it is fit by

$$E(\delta) = E_0 + \alpha\,\delta^2 + \beta\,\delta^4, \tag{1}$$

and the sign of the quadratic coefficient $\alpha$ is the whole story: while $\alpha > 0$ the centre is stable and the ion has one well; when $\alpha < 0$ the centre is unstable and the well has split in two. Figure 2 shows the crossover. At $c = 5.5$ Å, the dry spacing, $\alpha = +2.5$ eV/Å$^2$ and the well is single and stiff. By $c = 6.9$ Å, the monolayer-hydrate spacing, $\alpha = -0.5$ eV/Å$^2$ and the ion has a pair of wells about 0.7 Å off centre. The sign flips between two spacings we actually computed, so the critical spacing is bracketed by the data, 5.5 Å $< c^*_{\mathrm{Na}} <$ 6.9 Å; interpolating the four points puts it at $c^*_{\mathrm{Na}} \approx 6.40$ Å. Repeating the whole scan at the physical composition $x = 1/3$, in a $\sqrt{3} \times \sqrt{3}$ cell, moves the crossing by only about 0.2 Å. The transition is not an artifact of the full-coverage cell.

The electron gas turns on at the same place, and we can watch it two independent ways. The first is the density of states at the Fermi level, $N(0)$, computed from a dense $\mathbf{k}$ mesh: for the Na $1 \times 1$ series it is 0.03 states per eV per cell per spin at 5.5 Å and jumps to 1.06 at 6.9 Å, a factor of thirty, and then saturates. Below the transition the gallery is insulating; above it there are carriers. The second is the Bader charge that flows back onto the sodium as the gallery opens, $\Delta q = 0/0.14/0.40/0.58\ e$ across 5.5/6.9/8.4/9.9 Å; read as an areal density this is a few times $10^{13}$ carriers per cm$^2$, the scale of a real two-dimensional electron gas. The two probes are measuring different things—one the band edge crossing $E_F$, the other the occupation filling up—and they agree on when the gas appears. The soft mode and the gas appear together because they are one event: opening the gallery both unbinds the electrons from the oxygen and unbinds the sodium from the centre.

Figure 3 collects the three signatures on one axis, and it is the figure the rest of the paper points back at: the Landau coefficient $\alpha(c)$ crossing zero, the carrier density $N(0)$ switching on, and—built on top of them—the superconducting dome we come to next. Two warnings belong here, not later. First, beyond about 7 Å the "double well" is no longer truly a double well: the energy is still falling at the largest displacement we sampled, which means the sodium is sliding toward a bound site on one $CoO_2$ layer and the calculation has not reached the minimum. Those depths are lower bounds, and we mark them as such. Second, the well depths themselves carry a large systematic, discussed in Sec. VI, because we held the oxygen planes fixed. The sign of $\alpha$ does not; the single-to-double-well flip survives every check we ran.

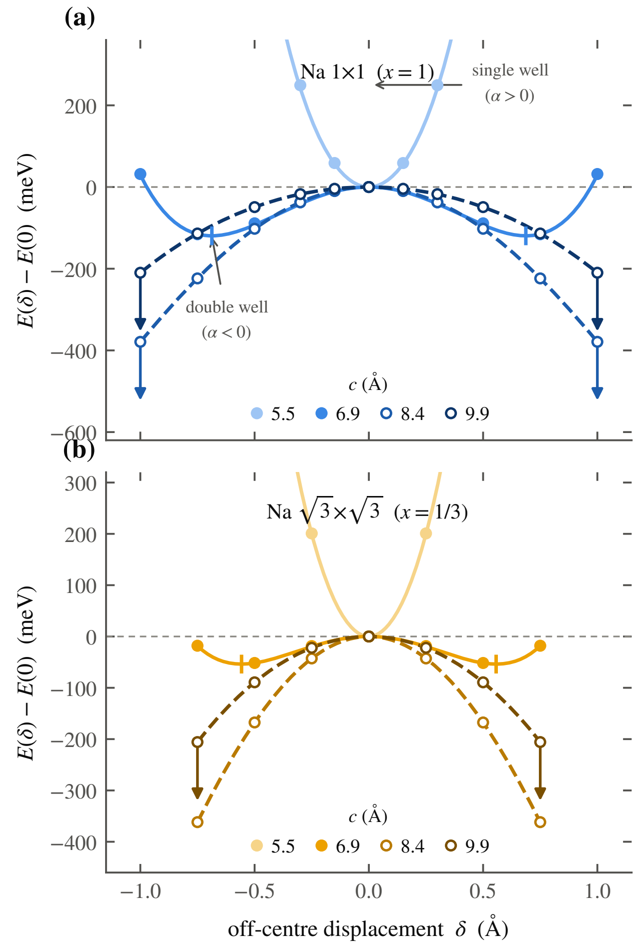


FIG. 2. The single-to-double-well transition. Energy $E(\delta)$ of the off-centre sodium displacement, computed for Na $1 \times 1$ (top, $x = 1$) and Na $\sqrt{3} \times \sqrt{3}$ (bottom, $x = 1/3$) at the four gallery spacings; points are total energies, curves are the fits of Eq. (1) continued to sixth order where needed. At 5.5 Å the well is single and stiff ($\alpha > 0$); by 6.9 Å it has split into a double well ($\alpha < 0$) with minima marked. Open symbols with arrows ($c \geq 8.4$ Å) flag scans that are still falling at the largest sampled displacement—there the sodium is adsorbing onto one layer and the quoted depth is only a lower bound.

## III. THE PAIRING WINDOW

Now we make the mechanism quantitative. We separate first what is borrowed from what is done here.

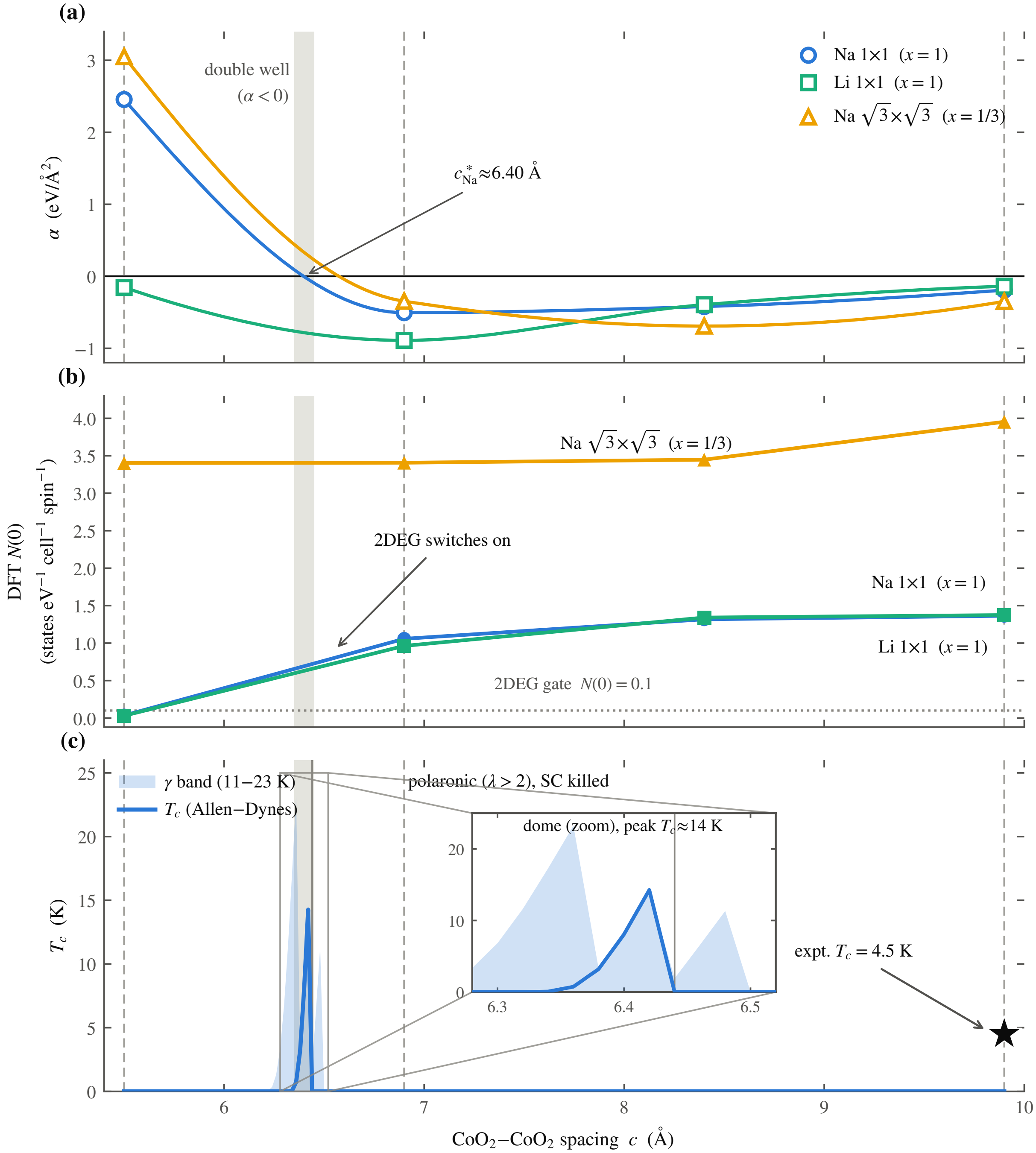


FIG. 3. The central result, three signatures of one transition against the $CoO_2$–$CoO_2$ spacing $c$. (a) The Landau coefficient $\alpha(c)$ for the three series; it crosses zero near $c^*_{\mathrm{Na}} \approx 6.40$ Å, where the single well becomes a double well. (b) The Fermi-level density of states $N(0)$, which jumps from essentially zero to of order one state per eV at the same spacing—the interlayer electron gas switching on. (c) The Allen–Dynes superconducting temperature: a narrow dome pinned just above $c^*$, with the shaded band spanning the electron–phonon coupling ansatz (11–23 K) and the thin vertical line at the $\lambda = 2$ crossing marking the polaron wall—everywhere to its right the coupling exceeds the polaron limit and superconductivity is killed. The inset magnifies the narrow dome. The experimental anchors are the thin dashed verticals at 5.5, 6.9 and 9.9 Å: the dry parent and monolayer hydrate (not superconducting) fall on the walls, and the bilayer hydrate (star, $T_c = 4.5$ K) does not sit in the static window—the difficulty of Sec. IV, resolved there by water re-binding the sodium.

The band structure is borrowed: the SSH-type four-band tight-binding model of Radha and Lambrecht [3] describes the two surface bands—one of alkali $sp_z$ character sitting in the gallery, one of $CoO_2$ character—that share a single electron according to how their centres are split. The splitting is set by the Bader charge we computed, so the gallery band fills exactly as the gas turns on in Fig. 3(b). What is done here is to let the sodium move and ask how the electrons feel it.

The coupling has a parity to it that decides the form of the answer. The alkali displacement $\delta$ is odd under reflecting the gallery through its midplane, so a linear electron–phonon coupling, which would shift the band energy in proportion to $\delta$, must vanish identically at the symmetric point. We checked this in the model at several $k_z$ and it holds. The leading coupling is therefore quadratic: the band energy responds as $\frac{1}{2}\chi\delta^2$, with

$$\chi = 2\sum_m \frac{|\langle \text{alk}|\,\partial H/\partial\delta\,|m\rangle|^2}{E_{\text{alk}} - E_m}, \tag{2}$$

the ordinary second-order perturbation sum, which comes out to $\chi \approx -6$ eV/Å$^2$ near the transition and grows as the band splitting collapses at wider spacing. Because the coupling is quadratic, an electron does not scatter off one quantum of the rattling mode; it scatters off two. The relevant matrix element is $\langle 0|\delta^2|2\rangle$ between the ground state and the second excited state of the quantized well—the first excited state is forbidden by the same parity—and the effective boson is the even overtone $\hbar\Omega = E_2 - E_0$. The dimensionless coupling is then

$$\lambda = \frac{2\,N(0)\,g^2}{\hbar\Omega}, \qquad g = \tfrac{1}{2}|\chi|\,\langle 0|\delta^2|2\rangle, \tag{3}$$

with $N(0)$ taken from the density-functional calculation, not fit.

The window has two walls, and the physics of each is simple. Close the gallery below $c^*$ and $N(0) \to 0$: there are no carriers, so $\lambda \to 0$ and there is nothing to pair. Open it too far, or cool the sodium into one well, and the ion freezes off-centre while $\hbar\Omega$ stays near 20 meV; the coupling then runs up past $\lambda \approx 20$, which is not strong superconductivity but a self-trapped polaron, and the pairing is killed. Between the two the quantized well is soft: the effective zero-point frequency of the sodium mode collapses from $\hbar\omega_{\text{eff}} = 30$ meV in the stiff single well to a fraction of a meV in the double well (Fig. 4), which is the soft, strongly anharmonic mode the mechanism needs. Feeding $\lambda(c)$, $\hbar\Omega(c)$ and $N(0)$ into the Allen–Dynes formula, with $\mu^* = 0.10$ and gated on $N(0) \geq 0.1$, gives the dome of Fig. 3(c): a narrow peak between $c \approx 6.35$ and 6.45 Å, of height $T_c \approx 14$ K at $\lambda \approx 1.3$ and $\hbar\Omega \approx 11$ meV. The dry parent (5.5 Å) sits on the low wall with no gas, and the monolayer hydrate (6.9 Å) sits past the high wall in the polaronic regime; both come out correctly non-superconducting.

We state the honest status of the number 14 K. The height of the dome depends on the electron–phonon range parameter as the fourth power, and over its plausible range the peak runs from about 11 to 23 K; the width, near 0.1 Å, depends on interpolating the well shape between four computed spacings and is not a precision statement. What is robust is qualitative and is the claim of this section: a dome exists, it sits pinned just above the gas turn-on and below the polaron crossover, its height is of the right order for a few-kelvin superconductor, and it contains neither the monolayer nor the bilayer anchor. The last of those is a problem, and we turn to it now.

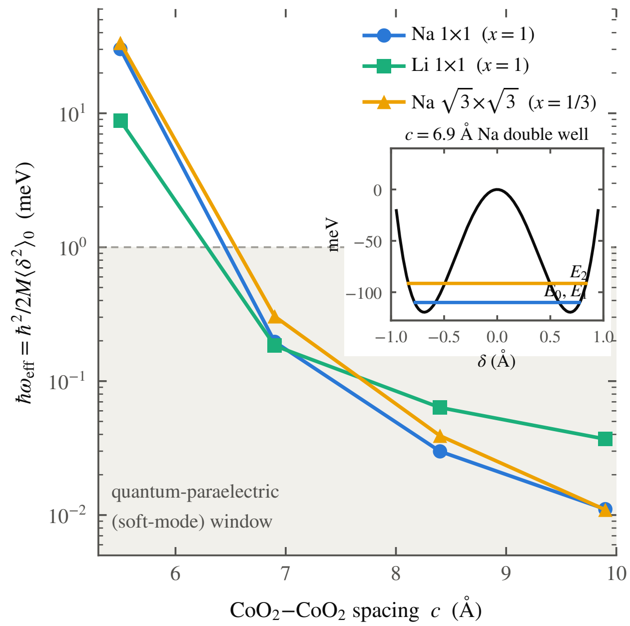


FIG. 4. The sodium mode softening. Effective zero-point frequency $\hbar\omega_{\text{eff}} = \hbar^2/2M\langle\delta^2\rangle_0$ of the quantized alkali mode, from an exact one-dimensional solution of the fitted wells at the physical masses. It collapses by three orders of magnitude across the transition, from a stiff $\sim 30$ meV single-well mode into a sub-meV soft mode as the gallery opens, entering a quantum-paraelectric window. Inset: the lowest levels in the 6.9 Å sodium double well, with the near-degenerate tunnelling doublet $E_0, E_1$ and the even overtone $E_2$ that carries the quadratic electron–phonon coupling.

## IV. THE DIFFICULTY AT 9.9 Å AND WHAT IT TEACHES

The bilayer hydrate is the compound that actually superconducts, and it holds its $CoO_2$ layers 9.9 Å apart [4]. That is far outside the dome we calculated. Worse, when we compute the sodium well at 9.9 Å without water, we do not find a soft symmetric rattler at all: the energy falls monotonically all the way to the largest displacement we sampled ($-206$ meV at $\delta = 0.75$ Å, still falling), meaning the sodium is not centred but sliding toward a bound site against one $CoO_2$ sheet. The curvature at the midplane is negative, $V''(0) < 0$: the centred ion is not the bottom of a soft well but the top of an instability, an imaginary-frequency mode ($|\hbar\omega| \approx 11i$ meV) that runs downhill into chemisorption. The static picture puts the one superconducting compound in the dead polaronic regime. This is not our artifact; it is the mechanical face of a result the electronic-structure literature settled long ago. The earliest water-inclusive calculation already found that hydration only makes the bands more two-dimensional [5]; Johannes and Singh put the negative statement sharpest, that intercalated water has "little effect . . . outside of the predictable effects of expansion" [6], and the same holds for the mono- and bilayer hydrates alike [7]. That static null result is the foil this calculation answers: a bare expanded gallery is a disaster for the sodium. The cause, we will show, is that these calculations hold the water still, and the water does not hold still [8]. The one earlier hint that the water's *motion*, and not its static presence, might matter came in a single sentence of Lynn and co-workers, who noted—by analogy to $MgB_2$—that anharmonic motion of the hydrogen and oxygen ions might enhance the superconductivity [9]; it was never developed into a mechanism, and developing it is the business of this paper.

So we put the water in. We built the bilayer solvation cage that neutron diffraction [4, 9] places around the sodium—four rigid gas-phase $H_2O$ molecules, two in the layer below the sodium plane and two above, octahedrally disposed about the ion (Fig. 5(b); the placement and its validation are in Appendix A)—and recomputed $E(\delta)$ in that cage, against the identical cell with the twelve water atoms deleted. The waters are frozen while the sodium scans, so the hydrate-minus-vacuum difference is clean. The result is Fig. 5(a), and it is the demonstration this paper turns on. The vacuum curve runs away as before. The hydrate curve does not: the solvation cage converts the runaway into a *bound* double well, with minima near $\pm 0.30$ Å, a central barrier of about 0.20 eV, and a real, confined $c$-axis sodium mode. Quantized about its true minimum with the sodium mass, that mode has $\hbar\omega_{\rm eff} \approx 23$ meV: finite, positive, bound—where the vacuum mode was imaginary. Water re-binds the sodium the static lattice was throwing away.

We are careful about what this does and does not settle. It settles the qualitative claim—the water is not a spacer, it is what keeps the sodium from chemisorbing, and it does so mechanically, not electronically. It does not, by itself, land the sodium in the dome. The cage here is frozen, and a frozen cage re-binds the ion into a still-deep well; feeding that well through the coupling machinery of Sec. III gives a large $\lambda$ (above the polaron limit across the whole ansatz range), because the barrier is 0.20 eV and the ion is localized, not yet rattling. The static half of the water's job—stop the chemisorption, re-symmetrize the potential—is now demonstrated. The remaining half—soften the well the rest of the way into the pairing window—is what the water's *motion* does, and a rigid cage cannot show it. This sharpens the claim rather than weakening it: it localizes the missing physics precisely, in the librations and hydrogen-bond fluctuations of the gallery water [8], and nowhere else.

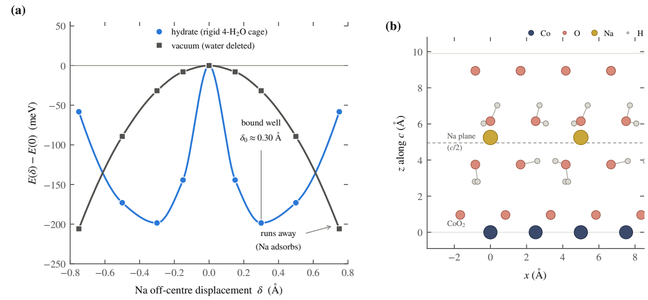


FIG. 5. The rigid-cage water demonstration. (a) Total energy $E(\delta)$ of the off-centre sodium at $c = 9.9$ Å in the $\sqrt{3}\times\sqrt{3}$ ($x = 1/3$) cell, with a rigid four-$H_2O$ solvation cage (hydrate) and with the water deleted (vacuum); points are QE total energies, curves guide the eye. Without water the curve runs away monotonically—the sodium chemisorbs onto a $CoO_2$ sheet, an unstable ($V''(0) < 0$) mode with no bound state. With the rigid cage it becomes a bound double well (minima $\approx \pm 0.30$ Å, central barrier $\approx 0.20$ eV), a real confined mode of $\hbar\omega_{\rm eff} \approx$ 23 meV. The frozen cage re-binds the sodium but leaves the well deep; softening it into the pairing window is the work of the water's motion (text). (b) The hydrate cell, drawn from the actual input coordinates (Appendix A), viewed along $c$: $CoO_2$ sheets at top and bottom, the sodium on its plane at $c/2$, and the two-layer water cage between.

The tests are therefore about dynamics. First, deuteration: replacing $H_2O$ by $D_2O$ stiffens the water librations by roughly $\sqrt{2}$ and tightens the hydrogen bonds, which should shift the sodium well and therefore $T_c$—a small, possibly inverse, isotope effect with an exponent well below the BCS one-half, because the boson is the sodium and not the water. We say plainly what our rigid-cage calculation can and cannot predict here: with the waters frozen, no atom that moves is deuterated, so the direct libration channel is absent by construction; a rigid run can bound only the sodium-mode mass dependence ($\hbar\omega \propto M^{-1/2}$) and the cage-stiffness mixing, and a genuine $D_2O$ number requires mobile water. Second, spectroscopy: there must be a soft, strongly anharmonic sodium $c$-axis mode at the scale we now quote, $\hbar\omega \sim 20$ meV, in the superconducting hydrate, visible to inelastic neutron scattering or low-frequency Raman, and it must be dynamical—sodium seen statically split, sitting off-centre without rattling, would kill the mechanism. Third, and decisive, the one calculation still open: $E(\delta)$ with the water allowed to move, by molecular dynamics or a relaxed-water potential of mean force. If mobile water does not soften the bound well of Fig. 5(a) toward the pairing window, the picture is wrong. The static calculation has answered the objection it was built to answer; the mechanism now stands or falls on the dynamical one.

## V. CONSEQUENCES

Table I lays the picture against the experimental record fact by fact. Here we take the consequences one at a time, and each is written so that it can fail.

Lithium cobaltate is the same structure with a smaller ion, and it has never been made to superconduct. Our picture requires it. Two things go against lithium at once. Its well crosses into the double-well regime only at the smallest spacing in our dataset, so we have only an upper bound $c^*_{\mathrm{Li}} \leq 5.5$ Å, and at that spacing the well is a mere 4 meV deep while the lithium zero-point energy is 8.8 meV—larger than the well itself. The light ion does not sit in either minimum; it is smeared across both. This is a quantum paraelectric, dynamically centred despite the negative $\alpha$, and a centred ion is a stiff ion with no soft anharmonic mode to offer. Lithium fails on the glue, and it fails for the plain reason that it is light.

The gas brings a magnetic phase with it, and that phase settles an old contradiction. Right where the gas first condenses, the gallery band is narrow and its density of states is high, and a narrow band with a high density of states is a Stoner magnet. The Radha–Lambrecht surface calculation already finds this polarized gallery gas [3], with the gallery moment antiparallel to the cobalt. In our picture the magnetism is therefore a thin strip hugging the carrier turn-on, strongest just past it and dying as the band widens—so superconductivity lives one step beyond the magnetism, and the magnetism lives one step beyond the turn-on. This is the ordering Sakurai, Ihara and Takada report for the magnetic phase bordering the dome against cobalt valence [10]. It also dissolves an old contradiction. Early powder NMR suggested a triplet, later single-crystal work a singlet; in this picture that is not a contradiction but a statement of where each sample sits, because the Stoner enhancement varies steeply across the strip and hydration drifts from sample to sample. The prediction is sharp: the NMR Knight-shift suppression below $T_c$ should anticorrelate, across samples, with the normal-state Stoner enhancement. Both symmetry-allowed states of Mazin and Johannes [2] are consistent with the muon null [11], and we do not need the chiral $d + id$ scenario [12] to fit the record, though our data do not exclude it.

One might worry that spin–orbit coupling selects or mixes the pairing. It does not, at any level that matters here. The gallery state is sodium-derived, with only a few percent cobalt weight, so its effective on-band spin–orbit scale is a couple of meV; the Rashba splitting from the off-centre sodium, estimated from the displaced charge, is about 0.8 meV, of order the gap but far below the Fermi energy, and the parity-mixed component of the gap it induces is at the sub-percent level. On the average centrosymmetric structure the up and down sectors carry opposite Rashba fields and largely cancel. Spin–orbit coupling is a small correction to the spin response, not a term in the pairing.

The gallery gas leaves a fingerprint on the sodium valence that two kinds of probe should read differently. If the sodium keeps a fraction $q$ of its electron in the gallery rather than giving it to the cobalt, then a site-specific valence probe—the cobalt $L$-edge or NQR—and a charge-counting probe—redox titration of the whole crystal—should disagree, and only when the gallery is open. We can now put a number on it. The Bader charge on the sodium in the $x = 1/3$ cell rises from the empty-gallery monolayer spacing to the superconducting bilayer by $\Delta q \approx 0.18\,e$, so the gallery-specific valence split is $\Delta v = x\,\Delta q \approx 0.06$ per cobalt, switching on with the water and vanishing in the monolayer hydrate where the gallery band is empty. We are careful here: this small split is not the large valence offset seen in these materials [13], which needs an extra electron reservoir such as oxonium and is several times larger; ours is the small, gallery-specific piece alone.

The picture points beyond this one compound. Three of its predictions we have now put to the calculation, and a fourth we leave for later. *Gating.* A $LiCoO_2$ thin film in which the carrier density is set electrostatically rather than chemically should show the gallery turn-on without the disadvantage of the light frozen ion. We added carriers to the anhydrous (5.5 Å) lithium cell by a compensating background, 0.1–0.3 $e$ per formula unit. The added charge fills the gallery: the Fermi level and the areal density rise monotonically (to $n_{2\mathrm{D}} \approx 4 \times 10^{14}$ cm$^{-2}$ at $0.3\,e$), and the lithium well softens as it fills—while remaining under a millielectronvolt deep, far below the lithium zero-point energy of $\sim 9$ meV, so the ion stays dynamically centred. Gating thus tunes the gas without freezing the ion, exactly the place to test the gas and the magnetism apart from the glue; it is the ionic-liquid-gating prediction for the thin-film collaboration. *Potassium.* A larger, heavier gallery ion should reach the soft-mode window at a wider spacing. It does: for K $1\times1$ the Landau coefficient crosses zero at $c^*_{\mathrm{K}} \approx 6.75$ Å, against 6.40 Å for sodium and $\leq 5.5$ Å for lithium, so the threshold orders Li<Na<K with ionic size, and the window opens farther out for potassium—which, hydrated to that spacing, should superconduct by the same route with a heavier, softer mode. *$Na_2CoSe_2O$.* The mixed-anion cobaltate [14] ($T_c = 6.3$ K) carries the same two-dimensional cobalt network with a different gallery chemistry, and our picture asks whether its gallery ion hosts an interlayer band. It does not: the states at the Fermi level are the Co $3d$–Se $4p$ network, and the ionic sodium of its $Na_2O$ gallery sits well above the Fermi level, with no sodium-derived band there. $Na_2CoSe_2O$ superconducts through its cobalt network, not the gallery, and is a clean negative control: same $CoO_2$-like sheet, no gallery gas, superconductivity by another route. Each is one experiment, and each is stated so it can fail.

### A. The pairing state through the same gauntlet

The pairing symmetry is not a question our mechanism leaves open. Twenty years of proposals for $Na_xCoO_2 \cdot yH_2O$ have been thinned by four measurements, and Table II sets the surviving field against them. Run our own state through the same gauntlet. The natural ground state of a phonon-mediated instability on the gallery gas is a spin singlet, anisotropic and $s_\pm$-like across the two bands—the gallery 2DEG and the $a_{1g}$ pocket—and it breaks no time-reversal symmetry. That single fact clears the two constraints that do the most damage in Table II: the singlet passes the single-crystal Knight shift that kills the triplet proposals, and the preserved time-reversal symmetry passes the $\mu$SR null [11] that kills the chiral $d + id$ states [12].

The other two constraints our picture turns to account. There is no Hebel–Slichter peak because the pairing boson is a soft, strongly anharmonic mode, and the anharmonic broadening that softens it also smears the coherence peak; we state this as expected of a soft-mode coupling, not as a fitted success. The specific-heat nodes are the more telling case. The gap on the gallery band is small—the dilute gallery density of states sets it—and a small gap on a two-dimensional gas is fragile against precisely the gallery disorder that dehydration introduces. A fragile second gap driven below the measurement threshold looks like line nodes. This is the one reading that explains both the node claims *and* their notorious dependence on sample age [10]: the gap is not nodal, it carries a soft second component that the aging of the water network quietly removes.

The old singlet-versus-triplet contradiction is then a statement of where each sample sat. Just past the carrier turn-on the gallery gas is near its Stoner boundary (Fig. 6), and there an equal-spin channel of $f$ symmetry lies close in energy to the singlet—the connection to the two $f$-wave states Mazin and Johannes could not eliminate [2]. A sample nearer the boundary leans triplet, one past it leans singlet, and hydration drifts samples across it. The state is falsifiable in two clean ways. Any breaking of time-reversal symmetry—a $\mu$SR relaxation onset at $T_c$, a polar Kerr rotation—excludes the singlet channel; so does a Knight shift that stays flat below $T_c$ in a fresh single crystal, because a singlet must show the shift drop. Either observation would end the picture.

### B. The design rule and its limits

If the picture holds it is a design rule: open a layered gallery past the spacing where the donor ion turns bistable, and the same motion that spills the ion's electrons into an interlayer gas softens the ion into the pairing boson, so that superconductivity occupies a window in spacing closed below by the carrier onset and above by polaronic freeze-out. Every ingredient of the rule is old. Pairing tuned to a soft polar mode at a quantum-critical point is the $SrTiO_3$ story [15]; an occupied interlayer state as the switch for superconductivity in an intercalate, set by the layer separation, is the graphite-intercalation story [16]; enhancement of $T_c$ as an intercalant mode softens, then collapse when the ion orders, is seen in $CaC_6$ under pressure [17]; and carriers in one layer paired by a quantum-paraelectric layer next door is an old model of Shenoy, Subrahmanyam and Bishop [18]. What is new is the joinery: that the donor ion is *itself* the quantum-paraelectric unit, not a caged spectator or a host-lattice mode, and that one structural knob—the gallery spacing—turns on the carriers and tunes the ion through its bistability at once, so the window closes on both sides rather than one.

A rule stated that broadly invites a counterexample, and the most-studied layered donor intercalate is one. In $Li_xZrNCl$ and its hafnium analogue [19], opening the gallery from 9 to more than 22 Å never kills the superconductivity: $T_c$ rises modestly and then saturates, and it survives with no intercalant ion at all, whether the chlorine is de-intercalated or the carriers are supplied by a gate [20]. This looks like a refutation until one checks the premise, which fails: here the donor's electron does not return to the gallery. It fills an in-plane Zr/Hf $d$–N $p$ host band, the ion is chemically locked and spectroscopically silent, and the pairing belongs to the host, not to any gallery gas or soft ion. The nitride is therefore orthogonal to our rule—it tests a different pairing channel—and its use to us is disciplinary. It forbids the universal statement: the rule holds only where the interlayer state lies at the Fermi level, the precondition made computable by Csányi and co-workers [16], and fails where the host band lies below it. And it fixes the null hypothesis. A gallery opened without a soft donor ion gives a monotonic, saturating $T_c(c)$; the nonmonotonic dome we compute for the hydrate is, against that background, the anomaly that demands the extra ingredient—the soft sodium—which the nitride does not have.

## VI. INADEQUACIES OF THE CALCULATION

This calculation cannot do several things, and the softest numbers in it are load-bearing, so we set down plainly which they are.

The largest systematic is that we held the oxygen planes fixed at one height for every spacing and displacement. Moving them by 0.06 Å shifts single-point energies by as much as the well depth itself, so every well depth here carries a systematic of about ±50%. This does not touch the sign of $\alpha$: the single-to-double-well flip survives ±50% scaling and survives the convergence checks, and the polaronic coupling in the frozen regime is an order of magnitude past the limit, so no conclusion turns on the soft depths. But the peak height of the dome does depend on them, and we do not claim 14 K to better than the factor of two already stated.

The rest of the list is shorter. Our transition spacing

TABLE I. The mechanism against the record. Six established facts about $Na_xCoO_2 \cdot yH_2O$ and what the soft-mode, charge-bifurcation picture makes of each: five are explained by the static calculation, and one—the superconductivity of the bilayer hydrate at 9.9 Å—is a prediction contingent on water re-softening the sodium well (Sec. IV). The one-line reason for each verdict is given.

| **Experimental fact** | **Our verdict** | **Why it follows** |
|---|---|---|
| Anhydrous (5.5 Å) is not SC *(Foo 2003)* | explained | $\alpha > 0$: a single stiff well ($\hbar\omega \approx 30\,\text{meV}$) with $N(0) \approx 0$—no interlayer gas, and nothing to pair. |
| Monolayer hydrate (6.9 Å) is not SC *(Foo 2003)* | explained | Past the window: $\lambda \approx 20 \gg 2$, so the sodium self-traps into a polaron and pairing is killed. |
| Bilayer hydrate (9.9 Å) is SC, $T_c =$ 4.5 K *(Takada 2003)* | prediction | Bare geometry is polaronic; a rigid water cage re-binds Na into a confined 23 meV well (Fig. 5), but full softening into the window needs mobile water—the one open calculation. |
| No superconducting Li analogue | explained | $c^*_{\text{Li}} \leq 5.5$ Å and the lighter mass make a stiffer mode; Li never enters the soft-mode window. |
| Magnetic phase borders the dome *(Sakurai 2015)* | explained | The local moment turns on exactly at the $\alpha < 0$ well transition (Fig. 6), one step in from the dome. |
| Superconducting $T_c$ is only a few K *(Takada 2003)* | explained | Allen–Dynes with the DFT $\omega_{\text{eff}}$ and $\lambda$ gives a peak $T_c \approx 14$ K (band 11–23)—the right scale. |

TABLE II. The pairing gauntlet. The leading proposals for the pairing state of $Na_xCoO_2 \cdot yH_2O$ set against the four measurements that have done most of the eliminating: the single-crystal $^{59}$Co Knight shift (it drops below $T_c$ in both field orientations, favouring a spin singlet), the absence of a Hebel–Slichter coherence peak in $1/T_1$, the low-temperature specific heat (a line-node $T^2$ term in some samples, two-gap fits in others, with an explicit sample-age dependence), and the zero-field $\mu$SR bound on broken time-reversal symmetry. Verdicts are read against each state's own symmetry; "reconciliation needed" marks a constraint a state meets only with an added assumption. The Sakurai entry is a mechanism, not a fixed symmetry, so the symmetry columns do not constrain it. The last row is the state of this paper (Sec. V A).

| **Proposed state (proposers)** | **Knight shift → singlet** | **No Hebel–Slichter** | **Specific-heat nodes, sample age** | **$\mu$SR: no TRSB** |
|---|---|---|---|---|
| RVB singlet, $d + id$ family (Baskaran [21]; Kumar–Shastry [22]; Wang–Lee–Lee [23]) | passes (singlet) | passes | tension (fully gapped) | fails (TRSB) |
| Spin-triplet $p/f$, $e'_g$ pockets (Tanaka–Hu [24]; Kuroki–Tanaka–Arita [25]) | fails (triplet) | passes | passes ($f$ nodes) | passes; needs $e'_g$ pockets ARPES does not see |
| Chiral $d + id$, fRG (Kiesel *et al.* [12]) | passes (singlet) | tension (fully gapped) | tension (fully gapped) | fails (TRSB) |
| Two surviving $f$-wave (Mazin–Johannes [2]) | reconciliation needed | passes | passes | passes |
| $s$-wave, anharmonic H/O phonons (Lynn *et al.*, aside [9]) | passes (singlet) | fails (predicts peak) | fails (fully gapped) | passes |
| Magnetic-fluctuation pairing (Sakurai *et al.* [10]) | symmetry not derived; the support is the phase diagram—a magnetic phase borders the dome (Table I) | | | |
| **This work**: singlet $s_\pm$ on the gallery 2DEG, TRS-preserving, two-gap | passes (singlet) | passes (anharmonic broadening) | passes; small 2DEG gap is disorder-fragile | passes |

is computed at full sodium coverage, $x = 1$, and only checked at the physical $x = 1/3$ in a single cell, where it moves by about 0.2 Å. The local-spin-density treatment over-polarizes the $\sqrt{3}$ cell, giving an unphysically large and flat moment there; we therefore rest the magnetic story on the $1 \times 1$ series and exclude the $\sqrt{3}$ moments, and we have checked that the well energies themselves are insensitive to the spin treatment at the few-percent level, so the over-polarization does not touch the wells. The explicit water we do include (Sec. IV, Appendix A)

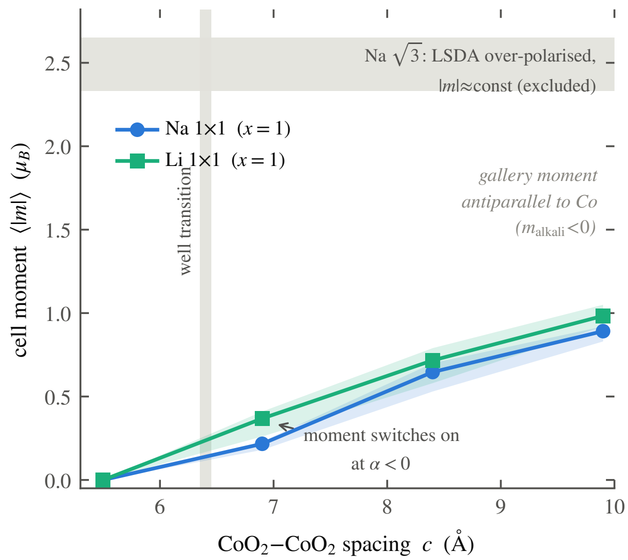


FIG. 6. The magnetic moment turning on with the well transition. Cell-integrated moment $\langle|m|\rangle$ for Na $1 \times 1$ and Li $1 \times 1$ against gallery spacing; markers are the mean over the displacement scan and the bands its spread. The local moment switches on at the same spacing where $\alpha$ turns negative, and the gallery moment is antiparallel to the cobalt. The $\sqrt{3}$ cell is over-polarized in LSDA and is shown only as an excluded caveat band.

is a rigid solvation cage, frozen while the sodium scans; it settles the static question—the sodium binds into a confined well rather than chemisorbing—but by construction it cannot show the softening that the water's motion supplies, which is the one calculation Sec. IV still leaves open. And the pairing is treated as a single Einstein boson in the Allen–Dynes approximation, with an explicit cut where the coupling leaves the regime that approximation can describe; a real bipolaron treatment past that cut is not attempted. The final test of any physical theory lies, of course, in experiment, and Sec. IV and Sec. V give the ranked list, the deuteration shift and the soft-mode search first.

## TODO — GENUINELY PENDING, TO BE INTEGRATED BEFORE SUBMISSION

- **TODO — Mobile-water $E(\delta)$ (Sec. IV)** — the one open calculation: $E(\delta)$ with the gallery water allowed to move (AIMD or a relaxed potential of mean force), to show the softening of the rigid-cage bound well of Fig. 5 the rest of the way into the pairing window. The rigid-cage static half is done (set G); this is the dynamical half.

## VII. SUMMARY

Sodium cobaltate superconducts only when it is wet. We have shown, by ordinary density-functional calculations, that opening the gallery between its $CoO_2$ sheets past a critical spacing does two things at once—condenses an interlayer electron gas and softens the sodium into a rattling double well—and that a phonon-mediated dome lives in the narrow window between the layers too close and the sodium frozen. The dry parent and the monolayer hydrate fall correctly outside that window. The one compound that superconducts sits, in the static calculation, on the wrong side of it, and that mismatch is the paper's real lesson: the water is not the spacer that holds the gallery open but the lubricant that keeps the sodium soft at a spacing where a rigid lattice would freeze it. The mechanism is testable, and the test that decides it is a calculation of the sodium well with the water put in and allowed to move.

### Appendix A: The rigid water cage

The water calculation of Sec. IV (set G) is built to isolate one thing: what a solvation cage does to the sodium potential, with everything else held fixed. We describe it in full because the result turns on the construction.

*Cell and electronic settings.* The host is the $\sqrt{3} \times \sqrt{3}$ ($x = 1/3$) $Na_xCoO_2$ cell at the bilayer spacing $c = 9.9$ Å ($a = 5.00$ Å), the same cell and the same spin-polarized PBE+PAW settings (60/480 Ry wavefunction/density cutoffs, Marzari–Vanderbilt smearing 0.02 Ry, $6 \times 6 \times 2$ **k** mesh) used for the anhydrous scans of Sec. II. The $CoO_2$ sheets and the oxygen heights are frozen at the values used throughout, so the water is the only thing added.

*Water geometry.* Each water is a rigid gas-phase molecule, O–H bond 0.9572 Å and H–O–H angle 104.52°. Four are placed per cell, in the arrangement neutron diffraction finds for the bilayer hydrate [4, 9]: the sodium sits on the Co-top $(0, 0)$ site on its plane at $z = c/2 + \delta$, and the four water oxygens lie in two layers 1.2 Å below and above that plane, on in-plane sites not occupied by the sodium—the two O-top images in the lower layer, toward which the sodium moves away, and the two other Co-top images in the upper layer. Over the whole displacement scan the sodium–water oxygen distances run 2.05–2.57 Å (lower layer) and 2.92–3.13 Å (upper), an octahedral-like first coordination shell. The hydrogens point away from the sodium, toward the nearest $CoO_2$ sheet and radially outward, with the molecular bisector tilted 50° off the $c$ axis. The orientations are chosen to maximize the smallest intermolecular contact: every water–water, water–framework and water–sodium distance stays $\geq 2.05$ Å across the whole $\delta$ grid. The two down-pointing and two up-pointing molecules cancel the layer $z$-dipole exactly, so the cage adds no net dipole field.

*The scan.* The waters are held fixed while the sodium scans $\delta$, so $E_{\rm hyd}(\delta) - E_{\rm vac}(\delta)$—the hydrate curve minus the identical cell with the twelve water atoms deleted—is the clean effect of the solvation cage on the sodium potential, with no relaxation confounds.

*What a rigid cage can and cannot say.* By construction this probes the *static* solvation cage only. It answers the objection it was built for—whether an expanded gallery leaves the sodium chemisorbed (it does, in vacuum) or bound in a confined well (it does, with the cage)—and it fixes the sodium-mode frequency of that well. It cannot represent the water's own motion: the librations and hydrogen-bond fluctuations that a mobile network supplies [8], and with them the direct deuteration channel and the further softening of the well, are absent here by design. That is the content of the one calculation we leave open.

## ACKNOWLEDGMENTS


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